

Infrared Line Tracking Data Collection

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Experimental Purpose

The Lastbit car is equipped with a four-channel infrared line tracking module at the factory. In this section, we will learn how to read the data returned by the four-channel line tracking module.

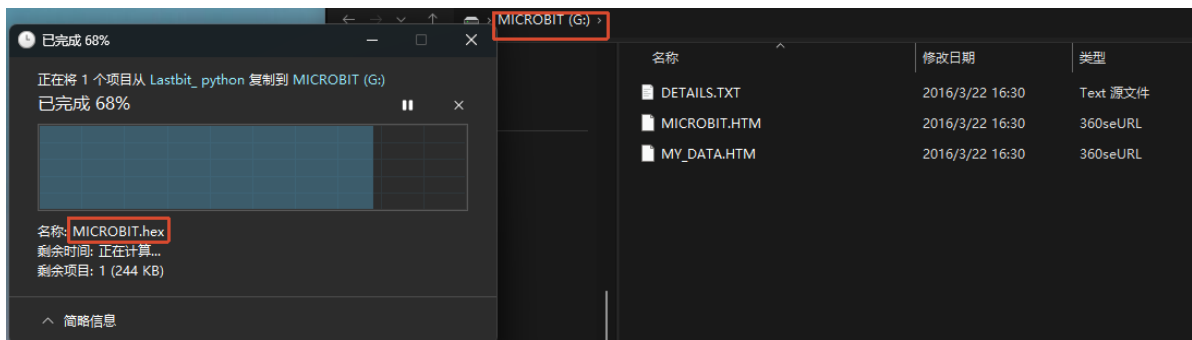


The infrared line tracking module consists of four pairs of infrared transmitting tubes and receiving tubes. During operation, the infrared transmitting tubes emit infrared light. The infrared light is reflected after shining on the surface of an object, and the receiving tubes output high or low level signals based on the intensity of the reflected light. Black objects absorb infrared light, so the reflection is weak and the module outputs a low level; white objects reflect more strongly, so the module outputs a high level. By detecting the level status of the four channels, the position of the car on the black line track can be determined.

The interface of the four-channel line tracking module has been connected at the factory, and the interface is designed with foolproof protection, so there is no need to worry about damaging the device when using the factory-provided cables.

Experimental Steps

Before flashing, make sure `LASTBIT.hex` has been flashed; otherwise, an error will occur and the module cannot be imported.



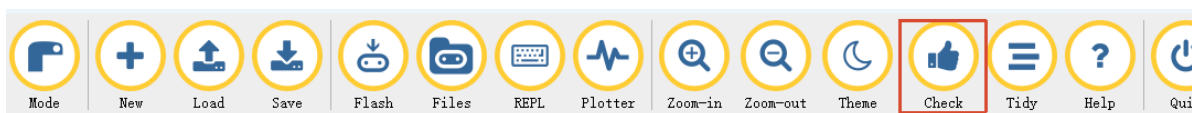
1. Before flashing, switch the switch to the OFF position and connect the computer to the microbit main board using a microUSB cable. **Note: it should be the Microbit interface, not the expansion board's Charging port.**
2. Open the Mu software. Here you can directly copy the program source code we provide; the operation steps are as follows. You can also enter the code yourself in the editor window. Note! All English characters and symbols should be entered in English input mode. Use the Tab key for indentation, and end the last line with a blank line.
Click the **New** button in the toolbar at the top left, and a file titled untitled will appear.



3. Copy the content of the **Code Implementation** section provided below into the current file. You will notice a small red dot after the title bar, indicating that the code content has been modified but is not yet saved.

Whether or not you save does not affect the code check and flashing.

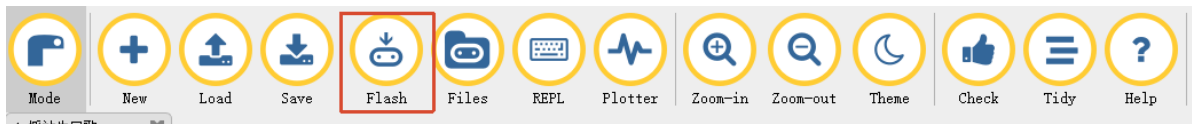
4. At this point, you can click **check** to check whether the code format has any problems.



5. If you have connected the MICROBIT to the PC in advance following the steps above and have flashed **LASTBIT.hex**, the next step is to download the current program to the Microbit.

Click the **Flash** tool in the toolbar to start flashing!

During flashing, the orange-yellow indicator light near the Microbit main board will blink frequently, and the editor will show **Copied code onto micro:bit.** at the bottom, indicating that the flashing is complete. After flashing, the indicator light will no longer blink.



6. Unplug the Microusb data cable and switch the switch to the ON position.

Code Implementation

```
# 3.Infrared_Line_Tracking_Data_Collection
```

```
from microbit import *
from lastbit import LASTBIT

Lastbit = LASTBIT()

while True:

    # Read track sensor data
    track_data = Lastbit.read_track()
    # If there is data
    # Read track sensor data
    if track_data is not None:
        left = track_data[0]
        mid_l = track_data[1]
        mid_r = track_data[2]
        right = track_data[3]

    # Clear the screen
    display.clear()

    # Show LED based on sensor status
    if left == 0:
        display.set_pixel(4, 2, 9)
    if mid_l == 0:
        display.set_pixel(3, 2, 9)
    if mid_r == 0:
        display.set_pixel(2, 2, 9)
    if right == 0:
        display.set_pixel(1, 2, 9)

    sleep(50)
```

Here, `left`, `mid_l`, `mid_r`, and `right` represent the data returned by the infrared line tracking module probes from left to right.

`read_track()`: reads the data of the four-channel tracking module. If a black line is detected, the corresponding blue light on the module lights up and returns 0; if it is white, it returns 1.

`display.set_pixel(x, y, value)`: used to light up the 5x5 LED matrix screen. The first parameter `x` indicates the column, the second parameter `y` indicates the row, and the third parameter `value` indicates the brightness, which can be 0 to 9.

Experimental Phenomenon

Lift the car up away from the desktop, and you can see that all 4 LED lights are on, as shown in the figure below.



At this time, the module returns data 0. Place a finger under one of the sensors, and the corresponding LED light will go out, and the returned data will be 1. You can test the four sensors one by one. The middle row of LEDs on the Microbit will light up or go out along with the indicator lights of the four-channel tracking module. If a certain sensor always stays in the state where the LED light is on, you can try rotating the potentiometer knob above the module to adjust the sensitivity.